Bihar Engineering University, Patna

B.Tech. 3rd Semester Examination, 2023

Course: B.Tech. *Time: 03 Hours*
Code: 101302 *Subject: Basic Electronics* *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 *Choose the correct answer of the following (Any seven question only):* [2 x 7 = 14]

(a) The average value of the full-wave rectified sine wave with period π and a peak value of V_m is
(i) $0.707V_m$ (ii) $0.5V_m$ (iii) $0.637V_m$ (iv) $0.318V_m$

(b) The most commonly used transistor configuration for use as a switching device is
(i) Common-base configuration (ii) Common-collector configuration
(iii) Collector-emitter shorted configuration (iv) Common-emitter configuration

(c) The Gunn diode is made of
(i) Silicon (ii) Germanium
(iii) Gallium Arsenide (iv) Selenium

(d) If for a silicon N-P-N transistor, the base – to – emitter voltage is 0.7V and the collector – to – base voltage is 0.2V, then the transistor is operating in the
(i) Active region (ii) Saturation region
(iii) Reverse active region (iv) cut-off region

(e) What current does $I=Aq(\frac{D_p}{L_p N_D}+\frac{D_n}{L_n N_A})n_i^2$ represent in pn junction diode?
(i) Forward current (ii) Diffusion current
(iii) Drift current (iv) Reverse saturation current

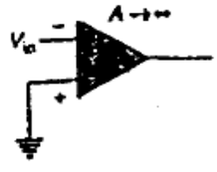
(f) For normal operation of a transistor
(i) Forward bias the emitter diode and reverse bias the collector diode
(ii) Forward bias the emitter diode as well as the collector diode
(iii) Reverse bias the emitter diode as well as the collector diode
(iv) Reverse bias the emitter diode and forward bias the collector diode

(g) Transistor is in saturation when
(i) $I_B=I_C$ (ii) $I_B>\frac{I_C}{\beta_{dc}}$ (iii) $I_B=0$ (iv) $I_B<\frac{I_C}{\beta_{dc}}$

(h) If a differential amplifier has a gain of 20,000 and CMRR = 80 dB, its common mode gain is
(i) 2 (ii) 1 (iii) ½ (iv) 0

(i) An operational amplifier possesses
(i) Very large input resistance and very large output resistance
(ii) Very large input resistance and very small output resistance
(iii) Very small input resistance and very small output resistance
(iv) Very small input resistance and very large output resistance

(j) If the input (V_{in}) to the circuit is a sine wave, the output will be
(i) Half-wave rectified sine wave (ii) Full-wave rectified sine wave
(iii) Triangular wave (iv) Square wave



Q.2 (a) With a neat circuit diagram and waveform, explain the working of bridge rectifier. [7]
(b) Explain Zener diode as voltage regular with no load.

Q.3 (a) Explain the operation of enhancement MOSFET.
(b) Draw the DC load line for transistor and identify Q point.

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Q.4 (a) With a neat circuit diagram, explain the working of wein bridge oscillator using op-Amp. [7]

(b) Explain common base input characteristic of BJT. [7]

Q.5 (a) An amplifier with open loop voltage gain A_f = 1000 ±100 is available. It is required to have amplifier whose gain varies by not more than ± 0.2%. Find [7]

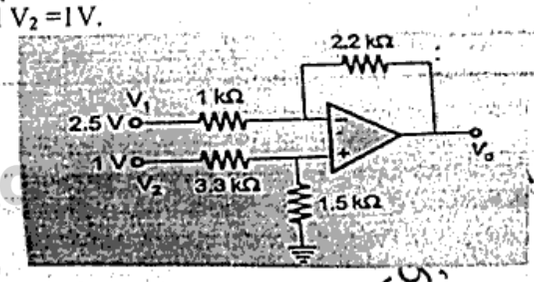
(i) Reverse transmission factor β of the feedback network

(ii) Gain with feedback.

(b) Define, four basic ways of connection the feedback signal. Such as voltage-series feedback, voltage-shunt feedback, current-series feedback, current-shunt feedback. [7]

Q.6 (a) Explain the differences between BJT & FET. [7]

(b) Calculate the output voltage V_0 of the circuit shown in figure. The input voltages are V_1 =2.5V and V_2 =1V. [7]



Q.7 Draw and explain pin configuration of 741 OP-Amp. [14]

Q.8 What is an SCR? Explain its working with switching characteristics. [14]

Q.9 Write short notes on *any two* of the following: [7x2=14]

(a) Differential amplifier

(b) Explain positive half wave rectifier with input and output waveforms

(c) Define OP-Amp parameter, Gain, CMRR, Slew rate, Input Resistance.

(d) Explain inverting amplifier.

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